PIC Delivery Plan

How we take the existing supplier-capability-twin codebase to an enterprise-grade AlphaHelion Purchasing Intelligence Cloud, sprint by sprint, against the Product Blueprint and the Development Technical Document.

VERSION	BUILDS ON	VERIFIED BY	CODEBASE
1.0 · 26 October 2026	AHG/PRC/PIC/2026-10/01 Blueprint · /02 Technical Document	AHG/PRC/PIC/2026-10/03 Test-Case Workbook	backend (FastAPI) · frontend (Next.js 16)

1. Summary

The current codebase covers most of the blueprint's 58 screens and 9 flows as a working demo. Only a thin core (PO, RFQ, invoice, payment, inventory, tenant) is stored in the database; the rest lives in process memory. There is no working authentication, no tenant isolation, no guarded state machines and no automated test suite. The business logic for the rules engine, SoD, lineage and the P2P flow is correct and will be carried forward.

This plan keeps the existing stack (FastAPI backend, Next.js frontend), adds the platform foundation the Technical Document requires, then rebuilds each module on that foundation in the order the blueprint's roadmap sets. Two phases are added to the blueprint's roadmap: a stabilisation sprint at the start and a hardening phase before general availability.

26	12 Mar 2027

two-week sprints, S0–S25	MVP procure-to-pay gate (end of S9)
22 Oct 2027 General availability (end of S25)	**27** engineers across six squads

MVP DEFINITION OF SUCCESS (FROM THE BLUEPRINT, §28)

- Touchless PO at least 30% for catalogue and contract lines in the pilot plant.
- First-pass invoice match at least 80%.
- Zero duplicate ERP postings across 10,000 invoices in UAT.
- Every PO line traceable to its origin in one click.

> **Team size.** Dates assume the six squads listed in the Technical Document (§18). With a team of about 8 engineers the order of work stays the same and the duration roughly doubles. Phase 0 cannot be shortened in either case.

2. Starting point

Findings from the codebase audit on 26 October 2026, each verified against the source.

AREA	CURRENT STATE	REQUIRED BY TECHNICAL DOCUMENT
Persistence	PR, sourcing, contracts, savings, MRP, MDM, approvals, SoD, outbox and lineage held in module-level dicts; lost on restart	PostgreSQL 16, module-owned tables
ORM setup	Three separate SQLAlchemy `Base` objects; Alembic sees one; `suppliers*` tables never migrated; app DB port/name differ from docker-compose	One metadata, one engine, all tables migrated

AREA	CURRENT STATE	REQUIRED BY TECHNICAL DOCUMENT
Money	29 `Float` columns, no currency	Integer minor units + ISO currency + FX rate and date
Authentication	`get_current_user` unused by any endpoint; `User` has no `hashed_password` column so register/login fail; login backdoor; frontend role guessed from email	SSO via OIDC/SAML, MFA for approvers, finance, admins
Tenancy	`tenant_id` from query string; many unscoped queries	RLS on every table; tenant from token only
State machines	Any PO status can be set directly; invoices can be paid twice; 3-way match creates its own goods receipt	Explicit transitions, guards, 409 on illegal transition
Rules	DMN-01/02/03 and PO risk score correct but hardcoded	Effective-dated rule sets; `rule_set_version` stored on documents
Events	Outbox is an in-memory dict with fake ERP numbers	Transactional outbox, Kafka, schema registry, idempotent consumers
AI	Keyword matching with canned answers	LiteLLM gateway, answer contract, autonomy guard, run log
Tests	Ad-hoc scripts against a live server; no frontend tests	Pytest/Jest 85% on domain logic, Pact, Testcontainers, Playwright, k6
Frontend	120 pages, about 30 fully hardcoded; `localhost:8080` in 24 files; three API calling styles; no guards, error pages, form validation or aria attributes	Typed API, object-page pattern, WCAG 2.1 AA, separate supplier portal

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3. Delivery decisions

These adapt the Technical Document's architecture decisions (AD-01 to AD-07) to the existing codebase. Each needs sign-off from the solution architect in Sprint 0 and is

recorded as an ADR in `docs/adr/`.

#	DECISION	REASON	CONSEQUENCE
DD-01	Python FastAPI modular monolith instead of NestJS services	Keeps about 23k lines of working logic; the document already specifies FastAPI for ML services	One deployable with strict module boundaries (no cross-module table access, enforced by import-linter). Modules can be split into services later without data changes.
DD-02	PostgreSQL 16 with RLS, SQLAlchemy 2.0, Alembic (AD-02)	As specified	App connects as a non-owner role; `SET LOCAL app.tenant_id` per transaction; RLS suite in CI
DD-03	Transactional outbox to Kafka-compatible broker (AD-03)	As specified	Redpanda locally and in dev; managed Kafka (Confluent or GCP Managed Kafka) in staging/prod; Avro schemas in a schema registry
DD-04	Camunda 8 for approvals and F1–F9 (AD-01)	Business-readable, versioned processes	Python job workers via `pyzeebe`. Confirm Camunda 8 Self-Managed production licensing in Sprint 0; fallback is SaaS.
DD-05	Keycloak as identity broker	Self-hostable OIDC/SAML with MFA, federates customer IdPs	Realm per environment; tenant and ABAC attributes issued as token claims
DD-06	Spend cube: Postgres materialised views in MVP, ClickHouse from Phase 2 (AD-04)	MVP volumes don't need ClickHouse; avoids early ops cost	CDC with Debezium into ClickHouse from S14
DD-07	LLM access only through the LiteLLM gateway (AD-06)	As specified	`litellm` is already a dependency; PII redaction and logging at the gateway
DD-08	Effective-dated rule sets in Postgres, evaluated in Python (AD-07)	Existing `rules_engine.py` logic stays; thresholds move to data	Camunda DMN used only where a decision gates a BPMN gateway

#	DECISION	REASON	CONSEQUENCE
DD-09	Supplier portal as a second Next.js app in a monorepo	Technical Document §16	Built from S13; until then the existing `/supplier/*` routes stay, behind supplier scopes
DD-10	Digital-twin features kept as a separate bounded context	Evidence, scoring, development, DFSC and ESG belong to the Supplier Capability Digital Twin, which PIC reads from (§12.3)	The `twin` module exposes capability, evidence level, free capacity and trust score; PIC writes back OTIF, PPM and invoice accuracy
DD-11	GCP asia-south1 (Mumbai) primary, asia-south2 (Delhi) DR; Terraform	India residency default	GKE, Cloud SQL for PostgreSQL (or AlloyDB), Secret Manager, Cloud KMS, Cloud Armor

4. Target code structure

A monorepo with one backend package per module. Each module owns its models, state machines, commands, queries and event handlers; other modules reach it only through its public service interface or its events.

```
pic/
├─ backend/
│  ├─ app/
│  │  ├─ core/            config.py · logging.py · telemetry.py
│  │  ├─ db/              base.py (Base, TenantScoped, Money, UUIDv7) · session.py (RLS) · types.py
│  │  ├─ platform/        auth · rbac_abac · sod · rules · audit (hash chain) · outbox · idempotency
│  │  │                   errors (problem+json) · numbering · workflow (pyzeebe) · lineage · org
│  │  ├─ modules/
│  │  │  ├─ spend/        M01    ├─ sourcing/  M05    ├─ order/      M08    ├─ savings/  M11
│  │  │  ├─ planning/     M02    ├─ supplier/  M06    ├─ receiving/  M09    ├─ ai/       M12
│  │  │  ├─ mdm/          M03    ├─ contract/  M07    ├─ payables/   M10    └─ admin/    M13
│  │  │  └─ requisition/  M04
│  │  │     each: models.py · states.py · commands.py · queries.py · api.py · events.py · rules.py
```

```
│   │   ├─ integration/     connectors/{sap_s4,tally,oracle,d365,odoo} · edi · einvoice_gst · punchout
│   │   ├─ twin/            supplier capability twin context (existing evidence/scoring/profile code)
│   │   └─ main.py
│   ├─ workers/             outbox_relay · connector_worker · zeebe_workers · scheduler · mrp_worker
│   ├─ migrations/          Alembic (single history)
│   └─ tests/               unit · api · integration (Testcontainers) · rls · sod · contract (Pact)
├─ apps/
│   ├─ web/                 buyer workspace (current frontend, refactored)
│   └─ supplier-portal/     from S13
├─ packages/
│   ├─ ui/                  design system: ObjectPage, DataTable, StatusChip, LifecycleStrip, ExplainPanel
│   └─ api-client/          generated from OpenAPI (openapi-typescript + openapi-fetch)
├─ processes/               BPMN/DMN models pic.mrp-to-pr … pic.material-request
├─ schemas/                 Avro event schemas
├─ infra/                   terraform/ · helm/ · docker-compose.dev.yml
└─ e2e/                     Playwright suites E2E-01…, k6 scripts
```

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5. Team & ownership

SQUAD	COMPOSITION	OWNS
Platform	1 lead, 3 backend, 1 DevOps/SRE	Tenancy and RLS, identity, RBAC/ABAC, SoD, rules, workflow, audit, outbox, CI/CD, observability, infra
Plan & Request	1 lead, 3 full-stack, 1 QA	requisition, mdm, planning (MRP), catalogue and punch-out
Source & Contract	1 lead, 3 full-stack, 1 QA	sourcing, contract, supplier (SLM) and twin link, savings, supplier portal
Order to Pay	1 lead, 3 full-stack, 1 QA	order, receiving, payables

SQUAD	COMPOSITION	OWNS
Integration	1 lead, 2 integration engineers	ERP connectors, EDI, GST e-invoice, reconciliation, data migration
Intelligence	1 lead, 2 ML engineers, 1 data engineer	spend cube, models, agent runtime, Copilot, MLflow, evaluation
Shared roles	Product owner, solution architect, UX designer, security engineer, delivery manager	Backlog, ADRs, design system and screen fidelity, threat models and pen-test liaison, release plan

6. Timeline

Each column is one two-week sprint starting on the Monday shown. Faded bars are support or lower-intensity work.



> **Holidays.** S4 covers 21 December to 1 January. Plan it at 60% capacity, or shift every later date by one week.

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7. Sprint plan

Each sprint lists its deliverables and, at phase ends, the exit test that must pass before the next phase starts. Screen IDs, flow IDs and principle IDs refer to the blueprint.

Phase 0 · Stabilise & foundation S0–S3 · 26 Oct – 18 Dec 2026

S0

26 Oct – 6 Nov

Stabilise the codebase

- Create the Git repository, branch protection, CODEOWNERS per module, PR template with the definition of done.
- Pin `requirements.txt`; add `python-jose`, `passlib[bcrypt]`, `pydantic-settings`, `uuid6`; remove the duplicate `sqlalchemy` line. Introduce uv or `pip-tools` lock file.
- Delete dead code: `inventory.py.bak`, unregistered `app/api/*`, broken `api/routers/inventory.py`, `core/db.py`, duplicate notifications import.
- Remove the login backdoor, hardcoded DB password and JWT secret; add `core/config.py` with pydantic-settings and `.env.example`.
- Stop `GET /finance/invoices` from seeding data; guard `pay_invoice` against unmatched or already-paid invoices as an interim fix.
- Developer stack in `infra/docker-compose.dev.yml`: Postgres 16, Redpanda + schema registry, Keycloak, Camunda 8 (Zeebe, Operate, Tasklist), Redis, Mailpit, OpenTelemetry collector, Jaeger.
- Pytest harness with in-process `TestClient` and Testcontainers Postgres; port the useful `verify_*.py` scenarios.
- CI v1: ruff, mypy, pytest, eslint, `tsc --noEmit`, gitleaks, pip-audit, npm audit.
- Frontend: `NEXT_PUBLIC_API_URL` replaces the 24 hardcoded URLs; move `<Shell>` inside `<Providers>`.
- ADRs DD-01 to DD-11 signed off; Camunda licence confirmed; ERP sandbox access requested (SAP S/4HANA, Tally).

S1

9 – 20 Nov

Data foundation and tenancy

- Single `Base`; `TenantScoped` mixin (UUIDv7 id, tenant_id, created/updated at and by, `row_version` for optimistic locking); `Money` composite (minor units + currency); FX rate and date on converted amounts.
- New Alembic baseline that reconciles all existing tables, converts Float money to minor units and supplier ids to UUID, with a data-migration script for the current demo data.
- RLS helper applied to every tenant table; application DB role without BYPASSRLS; `tenant_session` dependency issuing `SET LOCAL app.tenant_id`.
- RLS isolation suite: for every table, tenant A cannot read, update or insert tenant B rows. Runs in CI.
- Org model (ADM-01): tenant, legal_entity, company_code, plant, storage_location, purch_org, purch_group, cost_center, wbs_element, currency, factory_calendar.
- Per-tenant number ranges (REQ-, PO-, GR-, INV-) using gap-free sequences.
- RFC 9457 `problem+json` handler with `code`, `ruleRef`, `traceId`; mappings for `StaleDataError`→412, illegal transition→409, `SodViolation`→409 `SOD_CONFLICT`.
- `Idempotency-Key` dependency for every POST (24 h replay, request-hash check); ETag/If-Match on updates.

S2

23 Nov – 4 Dec

Identity, authorisation, SoD and audit

- Keycloak realms; OIDC JWT verification (RS256, JWKS cache); `Principal` with tenant, roles, scopes and ABAC attributes (company codes, plants, categories, value limit). Tenant comes only from the token.
- Scope catalogue per module (e.g. `po:create`, `po:release`, `invoice:post`); `require(scope, abac=…)` dependency on every endpoint; an automated check fails CI if any route lacks it.
- The 14 blueprint roles seeded as Keycloak roles mapped to scopes; MFA required for approver, finance and admin roles.
- SoD engine on the database: `sod_rule` (tenant-configurable), `action_log` (who did what on which object), check at command time for humans and agents. Rule IDs re-aligned to blueprint SOD-01…07.
- Append-only `audit_event` with hash chain per tenant; INSERT-only grant; nightly chain verifier; `GET /v1/audit/{objectType}/{id}`.
- Frontend: Auth.js OIDC login, `middleware.ts` route guards by scope, generated typed API client in `packages/api-client`, 401 handling. Persona switch removed outside dev.
- ADM-04 Roles & SoD and ADM-06 Audit Explorer wired to real data.

S3

7 – 18 Dec

Rules, workflow, events, lineage, design system

- `rule_set` with scope tuple (tenant, company code, plant, category, purchase type), version and effective dates; most-specific resolver. DMN-01, DMN-02, DMN-03, PO risk score and quote policy moved from constants to rule-set data. ADM-02 editor writes new versions; documents store `rule_set_version`.
- Camunda 8 integration: `pyzeebe` worker framework, process deployment from `processes/`, human-task API backing the unified approval inbox, SLA timers and escalation.
- Transactional outbox: `outbox_event` written in the same transaction as the domain change; relay worker (`SELECT … FOR UPDATE SKIP LOCKED`) publishing to `pic.<tier>.<event>`; Avro event envelope (tenantId, eventId, occurredAt, actor, objectRef, version, traceId).
- `integration_message` with `UNIQUE(tenant, target_system, idempotency_key)`; connector worker framework with retry (exponential backoff, max 8), dead letter and replay (`POST /v1/integration/messages/{id}/replay`); SAP mock via WireMock.
- `document_link` table and lineage query; `lineage_service.py` logic moved onto it.
- OpenTelemetry traces across web, API, workers and Zeebe; structured JSON logs; `/healthz`, `/readyz`.
- Design system in `packages/ui`: ObjectPage (header, status chip, tabs Lines · Approvals · Documents · Lineage · Audit, action bar), DataTable with cursor pagination, StatusChip, LifecycleStrip, ExplainPanel, form kit (react-hook-form + zod), toasts, `error.tsx`/`loading.tsx`/`not-found.tsx`.

Exit test. A test document runs end to end: DMN-01 resolves the approver chain, Camunda creates the task, an approver completes it, SoD blocks a conflicting approver, audit chain verifies, the event reaches the broker once. RLS suite green.

Phase 1 · MVP procure-to-pay S4–S9 · 21 Dec 2026 – 12 Mar 2027

S4

21 Dec – 1 Jan

Requisitions (M04)

- Tables requisition, pr_line, account_assignment, budget, budget_commitment; PR state machine (Draft → Submitted → In Approval → Approved → Partially Ordered → Ordered → Closed | Rejected | Returned | Cancelled).
- Purpose code mandatory (P01); origin stored and immutable via DB trigger (P02); free-text lines with `CHECK (item_type <> 'FREE_TEXT' OR material_id IS NULL)` (P04); purchase type per line driving the P08 matrix.
- Budget check at submit: hard stop for capex and WBS, soft stop with justification for opex; commitment on approval.
- F2 `pic.requisition` with DMN-01 approver chain and 2-business-day SLA. REQ-02 form and REQ-03 approval inbox rebuilt on the design system.

S5

4 – 15 Jan

Guided buying and material governance (M04, M03)

- REQ-01 search enforcing contract item → catalogue item → material → free text (P03); Postgres full-text and trigram search until OpenSearch lands.
- Catalogue and catalogue_item; quote policy by value band; PR change re-approval only when value, account or supplier changes. REQ-04 buyer workbench with origin filters, bundling and assign.
- MDM-01 material request with mandatory classification; duplicate score (description similarity, MPN, attributes, classification): ≥ 0.92 blocks, 0.75–0.92 warns. Promote free-text line to material request; PR keeps its line.
- F9 `pic.material-request` with parallel steward tasks by domain; MDM-02 steward workbench; SOD-06.

S6

18 – 29 Jan

Purchase orders (M08)

- Tables purchase_order, po_line, po_schedule_line, po_version, po_confirmation, info_record, source_list; DDL as Technical Document §19.2.
- PO from PR with source determination (info record and source list now; contract and quota in S12/S15) and inheritance of PR, price, Incoterms, payment terms and tolerances (P07). Overrides require `deviation_reason`.
- PO validation: risk score with explainable factors, DMN-02 touchless check, split-PO detection (same supplier + material group + cost object within 7 days), blocked-supplier guard, SOD-01/03/04.
- F5 `pic.po`; transmission by portal and email; supplier confirmation; change after confirmation creates version n+1 and requires re-confirmation.
- PO-01 workbench and new `/purchase-orders/[id]` object page (PO-02) with Lineage and Audit tabs; PO-03 confirmations.

S7

1 – 12 Feb

Receiving and supplier master basics (M09, M03, M06)

- ASN, goods_receipt, gr_line, inspection_lot, usage_decision, return_delivery; quality stock until usage decision; over-delivery above tolerance blocked; batch/serial/expiry capture; GR reversal.
- Service entry sheets (milestone, T&M, unit-rate, progress %) approved by someone other than the supplier and the PO creator.
- Existing inventory module re-based on the new GR movements (stock remains the source for MRP on-hand).
- F6 `pic.receipt`; RCV-01 to RCV-04; supplier ASN in the portal.
- Supplier status per category (Registered → … → Blocked) enough to drive PO guards; MDM-04 bank-detail change with two finance approvers, call-back verification, cooling-off period and alert on change + invoice within 30 days (SOD-02).

S8

15 – 26 Feb

Invoices and ERP posting (M10)

- Capture by portal, PDF upload (OCR adapter) and GST e-invoice with IRN and signed QR validation.
- Duplicate control: unique (tenant, company code, supplier, supplier invoice no., fiscal year) plus fuzzy check (amount ± 1, date ± 3 days).
- Per-line match in Decimal: 3-way against accepted GR minus already invoiced, 2-way where GR not required, service invoices against approved SES; DMN-03 tolerances from the rule set; exceptions with type, delta and suggested action.
- Invoice state machine with payment block until all exceptions are resolved; credit notes; GRIR accrual.
- Posting only through the outbox with key `tenant|company_code|supplier|invoice_no`; SAP S/4HANA connector (OData, IDoc fallback) and Tally connector (XML via on-prem agent); query-before-retry on timeout; nightly reconciliation.
- F7 `pic.invoice`; INV-01, INV-02 (side-by-side PO · GR · invoice), INV-03 posting and reconciliation; ADM-05 integration monitor.

S9

1 – 12 Mar

MVP hardening and pilot readiness

- SPD-01 spend overview on Postgres materialised views; INV-04 payables insights.
- Legacy endpoints for PO, invoice and payment retired; all P2P pages on `/v1` commands.
- Playwright E2E-01 (manual PR path to posted invoice); idempotency soak of 10,000 invoices with injected timeouts and duplicates.
- k6 baseline (P95 < 400 ms on transactional APIs), ZAP DAST, RLS and SoD suites, WCAG check on P2P screens.
- Migration rehearsal 1 with pilot data (suppliers, materials, open POs); UAT with the pilot plant.

Exit test. E2E-01 passes; zero duplicate ERP postings across 10,000 invoices; first-pass match and touchless rates measured in UAT against MVP targets; every PO line opens its lineage in one click.

Phase 2 · Source & contract S10–S14 · 15 Mar – 21 May 2027

S10

15 – 26 Mar

Sourcing events (M05)

- sourcing_event, event_line, invitee, scoring_model, criterion, bid, bid_line, clarification; event types RFI, RFQ, RFP, reverse auction, multi-round, lot bidding.
- Events start only from approved requirements (PRs, MRP demand or category plan); invitees limited to suppliers Qualified for the category or Conditional with waiver (P06).
- Weights must sum to 100 and lock at publish. Bids sealed with envelope encryption (Cloud KMS per tenant); early access blocked and logged; unsealing at close writes `bids.unsealed`.
- F3 `pic.sourcing-award` close timer; SRC-01 cockpit and SRC-02 event builder (replacing the mock `/rfqs/create`); supplier bidding in the portal.

S11

29 Mar – 9 Apr

Evaluation and award

- Bid normalisation and TCO (price, freight, duty, packaging, carrying cost, quality cost, payment-term value); should-cost parametric baseline by commodity.
- Award scenarios: lowest cost, best value, split/dual, capacity-constrained MIP using OR-Tools with free capacity read from the twin.
- SRC-03 bid comparison, SRC-04 scenario studio, SRC-05 award approval with written rationale for non-lowest awards, conflict-of-interest declaration, threshold approvals; SOD-05.
- Award hands over to contract (S12) or PO.

S12

12 – 23 Apr

Contract lifecycle (M07)

- contract, contract_line, clause_library with fallback positions and risk scores, template, obligation, price_validity, consumption.
- Contract state machine; legal + finance approval above threshold; e-signature adapter (DocuSign or an Indian Aadhaar eSign provider).
- Activation publishes price validity to source determination; indexation (e.g. quarterly LME ± 3%) creates new validity periods and never overwrites.
- PO price must equal contract price within tolerance or the PO is flagged; consumption alerts at 80/90/100%; F4 with renewal timer at −120 days.
- CLM-01 workspace, CLM-02 clause review (redline agent placeholder until S18), CLM-03 consumption and compliance, CLM-04 obligation tracker.

S13

26 Apr – 7 May

Supplier lifecycle and portal (M06)

- Qualification per category with questionnaires, certificates, audits and sample approval (PPAP); certificate expiry timers move status to Conditional automatically.
- Blocking stops new POs and flags open POs for buyer decision; scorecards (OTIF, PPM, responsiveness, invoice accuracy, ESG) recompute monthly and on every GR/invoice event.
- Risk monitor with sanctions and financial-risk adapters; CAPA and development plans.
- Twin integration: read capability, evidence level, free-capacity band, trust score; write back OTIF, PPM, invoice accuracy.
- F8 `pic.supplier-onboarding`; SLM-01 Supplier 360 (replacing the mock page), SLM-02, SLM-03, SLM-04 (replacing the seven mock `/risk/*` pages); supplier portal app (SLM-05) with supplier scopes.

S14

10 – 21 May

Savings and spend cube (M11, M01)

- savings_initiative and savings_ledger; baseline methods fixed at creation; lifecycle Identified → Negotiated → Contracted → Realising → Validated | Lost.
- Realised savings only from `invoice.posted`; cost avoidance never counts towards budget; SOD-07 finance validation. SAV-01 register, SAV-02 waterfall.
- Debezium CDC into ClickHouse; spend cube dimensions and measures from blueprint §7.2; SPD-02 to SPD-06 including maverick rules with owner and disposition; OpenSearch for global search.
- Migration rehearsal 2 (open contracts, 24–36 months of spend history).

Exit test. RFQ-2231 awards 60/40 at weighted 400.8, activates SA-5520, and PO-77410 inherits price 398, N60 and FCA Trichy. Savings register shows SV-0412 negotiated 0.99 Cr/yr.

Phase 3 · Plan & predict S15–S19 · 24 May – 30 Jul 2027

S15

24 May – 4 Jun

MRP engine (M02)

- Planning parameters, demand elements (forecast, sales orders, reservations, WBS demand), factory calendar, quota arrangements.
- Netting, lot sizing (exact, fixed, EOQ, period, min/max, pack rounding) and lead-time scheduling as pure, property-tested functions (Technical Document §9.1).
- Run modes net-change, regenerative and simulation (results only, no documents); exceptions PAST_DUE, BELOW_SAFETY, RESCHEDULE_IN/OUT, CANCEL.
- Firm and convert to PR with source determination including contract and quota; ERP planned-order import adapter for tenants running MRP in the ERP.
- MRP-01 run console, MRP-02 planner workbench, MRP-03 exception monitor.

S16

7 – 18 Jun

MRP scale and forecasting

- Partitioned MRP workers to meet 50,000 materials in under 10 minutes; F1 `pic.mrp-to-pr`.
- Demand forecast model (gradient boosting + seasonal baseline), MLflow registry, MRP-04 forecast review with override reason.
- Spend classification model (text classifier + embeddings); below 0.80 confidence goes to the steward queue; SPD-09 spend forecast.

S17

21 Jun – 2 Jul

Agent runtime (M12)

- Agent registry (purpose, input events, tool scopes, default autonomy, value/category limits, run-as role).
- Autonomy guard before every tool call: role permission, SoD, autonomy level for tenant and category, value limit, kill switch. Denied calls become L2 drafts.
- Run log with inputs, evidence ids, model and prompt version, output, confidence, action, feedback; auto-suspend when acceptance falls below threshold or drift is detected.
- LiteLLM gateway with PII and bank-data redaction and per-tenant retention.
- First agents: Guided Buying (L2), PO Validation (L3), Invoice Exception (L2), Spend Classification (L3).

S18

5 – 16 Jul

Copilot and command centre

- AIX-02 Copilot returning the §13.3 answer contract, validated by JSON schema; tenant-scoped, permission-filtered retrieval (pgvector) before the model sees text; numbers computed by SQL and cube tools.
- AIX-01 command centre on real KPIs; AIX-03 agent console with autonomy settings and kill switches.
- Agents: Touchless PO (L4 within policy), Bid Analysis, Contract Review (redline vs playbook), Material Dedup, MRP Exception, Expediting with late-delivery model.

S19

19 – 30 Jul

Remaining agents and AI evaluation

- Demand Sensing, Sourcing Strategy, Supplier Risk, Savings Validation agents.
- Offline evaluation sets per model (classification accuracy, MAPE, answer-contract completeness) gating any autonomy increase; model cards.

Exit test. MR-1043 nets 12,000 – 1,400 – 1,200 – 600 + 200 = 9,000 and creates PLO-55821; six agents in production at L1–L3; autonomy guard tests prove no agent exceeds its role, SoD or value limit.

Phase 4 · Industry packs S20–S23 · 2 Aug – 24 Sep 2027

S20

2 – 13 Aug

Template engine, manufacturing and general enterprise

- Versioned seed bundles (purchase types, number ranges, event and contract templates, approval matrix, KPIs, demo data) applied at tenant creation and editable after.
- Manufacturing: PPAP/IATF gates, quota arrangements, scheduling agreements, ERS, metal-index clauses.
- General enterprise: punch-out (cXML PunchOutSetupRequest/OrderMessage, SAP OCI), P-card limits, receipt-less below threshold.

S21

16 – 27 Aug

Construction and IT services

- Construction: WBS on every line with hard budget stop, BOQ, retention and advance recovery, variation orders as contract amendments, progress-based SES.
- IT services: MSA + SOW, rate cards, timesheet and milestone SES, licence entitlement and auto-renewal alerts.

S22

30 Aug – 10 Sep

Logistics and healthcare

- Logistics: lane tenders and spot auctions, rate contracts with fuel indexation, POD as service confirmation, freight audit against lane rates.
- Healthcare: lot/expiry/serial and cold chain at GR, consignment with usage-triggered POs, recall traceability, clinical sign-off in the award.

S23

13 – 24 Sep

Remaining connectors

- Oracle Fusion/EBS (REST/FBDI), Dynamics 365 (OData/Dataverse), Odoo (JSON-RPC); connector SDK documentation.
- EDI gateway (ORDERS, ORDCHG, ORDRSP, DESADV, INVOIC over AS2/SFTP); e-way bill on ASN.

Exit test. Each industry template passes its test sheet in the workbook, including the demo case listed in blueprint Table 23.

Phase 5 · Hardening & GA

S24–S25 · 27 Sep – 22 Oct 2027

S24

27 Sep – 8 Oct

Production readiness

- Load test at 10,000 concurrent users per tenant and 5 M invoices/year volume; spend cube under 3 s on 50 M lines.
- DR drill against RPO 15 min / RTO 4 h; blue/green deployment rehearsal; backward-compatible migration check.
- Third-party penetration test against OWASP ASVS L2; WCAG 2.1 AA audit; DPDP privacy review; runbooks for dead-lettered postings, workflow incidents, MRP failure, model drift.
- Migration rehearsal 3 (full cutover dry-run).

S25

11 – 22 Oct

Cutover and general availability

- Pilot tenant production cutover following §15; open-invoice freeze window; reconciliation sign-off by finance.
- Two weeks hypercare with daily SLO review (PR approval latency, touchless rate, match rate, posting success).

Exit test. All release gates in §14 green; finance signs off reconciliation; GA announced.

8. Module specifications

What each module must contain when done. "Screens" maps each blueprint screen to the existing page it replaces; pages marked new do not exist yet.

M01 Spend & Category Intelligence `Intelligence squad · S9, S14, S16`

TABLES	spend_line, taxonomy, taxonomy_node, classification_review, maverick_rule, maverick_hit, category_plan, category_budget; ClickHouse `spend_cube` (Postgres materialised view in MVP)
APIS	`GET /v1/spend/cube`, `GET /v1/spend/maverick`, `POST /v1/spend/maverick-hits/{id}/disposition`, `/v1/category-plans`
EVENTS	Consumes invoice.posted, purchase_order.released
RULES	De-duplicate PO and invoice through lineage before aggregation; classification below 0.80 to steward queue; intercompany excluded from addressable; maverick rules configurable, each hit needs owner and disposition
SCREENS	SPD-01, SPD-06 ← `/spend/cube` · SPD-07 ← `/workbenches` (Kraljic) · SPD-08 ← `/savings` · SPD-02 to SPD-05, SPD-09 new
TESTS	Spend total ties to GL within ±0.5%; cube query under 3 s on 50 M lines

M02 Demand Planning & MRP `Plan & Request · S15–S16`

TABLES	material_plan_params, demand_element, forecast, mrp_run, mrp_run_result, planned_order, mrp_exception, quota_arrangement, factory_calendar
STATES	Planned order: Created → Firmed → Converted | Deleted (by next run if not firmed)
APIS	`POST /v1/mrp/runs`, `GET /v1/mrp/runs/{id}/exceptions`, `POST /v1/planned-orders/{id}/firm`, `POST /v1/planned-orders/{id}/convert`
EVENTS	mrp.run.completed, planned_order.firmed
RULES	Only MRP-active materials planned; firmed orders never changed; simulation creates no documents; conversion inherits material, qty, date, plant, storage location, source
SCREENS	MRP-01 to MRP-03 ← `/mrp/planner-workbench` · MRP-04 new
TESTS	MR-1043 = 9,000; property tests for lot sizing; 50,000 materials under 10 min

M03 Material & Supplier Master Data Governance `Plan & Request · S5, S7`

TABLES	material, material_request, duplicate_candidate, steward_task, supplier_master_change, bank_account (field-encrypted)
STATES	Draft → Duplicate Check → In Stewardship → Approved → Synced to ERP | Rejected | Merged
APIS	`POST /v1/materials/requests`, `POST /v1/materials/requests/{id}/decision`, `POST /v1/requisitions/{id}/lines/{n}/promote-material`, `POST /v1/suppliers/{id}/master-changes`
EVENTS	material_request.approved, supplier.master.changed
RULES	Duplicate ≥ 0.92 blocks unless steward overrides with reason; 0.75–0.92 warns; bank change needs two finance approvers, never the requester, plus call-back and cooling-off; ERP sync only after final approval; SOD-02, SOD-06
SCREENS	MDM-01, MDM-02 ← `/mdm/requests`, `/mdm/studio` · MDM-04 ← bank change in `/mdm/studio` · MDM-03 new
TESTS	Dedup precision/recall on labelled set; bank-change fraud scenarios from the workbook

M04 Requisitioning & Guided Buying `Plan & Request · S4–S5, S20`

TABLES	requisition, pr_line, account_assignment, budget, budget_commitment, catalogue, catalogue_item, punchout_session
STATES	Draft → Submitted → In Approval → Approved → Partially Ordered → Ordered → Closed | Rejected | Returned | Cancelled
APIS	`POST /v1/requisitions`, `POST /v1/requisitions/{id}/submit`, `/approve`, `/return`, `GET /v1/guided-buying/search`
EVENTS	requisition.submitted, requisition.approved
RULES	P01–P04, P08; DMN-01; quote policy (< ₹25,000 none; to ₹2 L one quote or catalogue; > ₹2 L three quotes or sourcing event); budget hard/soft stop; re-approval on value, account or supplier change
SCREENS	REQ-01 ← `/p2p` guided buying · REQ-03 ← `/approvals` · REQ-04 ← `/purchase-orders/pending` · REQ-02 new form
TESTS	Search-order enforcement; budget stop variants; DMN-01 band edges (24,999 / 25,000 / 2,00,000 / 10,00,000)

M05 Strategic Sourcing `Source & Contract · S10–S11`

TABLES	sourcing_event, event_line, invitee, scoring_model, criterion, bid (ciphertext + key ref), bid_line, clarification, award_scenario, award_decision, conflict_declaration
STATES	Draft → Published → Open → Closed → Under Evaluation → Award Pending → Awarded | Cancelled
APIS	`POST /v1/sourcing/events`, `/publish`, `/bids`, `/scenarios`, `/award`
EVENTS	sourcing.event.published, sourcing.bid.submitted, sourcing.event.awarded
RULES	Only qualified invitees; sealed until close; weights sum to 100 and lock at publish; non-lowest award needs rationale; above threshold category head + finance; SOD-05
SCREENS	SRC-01 ← `/sourcing` · SRC-02 ← `/rfqs/create` · SRC-03 ← `/rfqs/compare` · SRC-05 ← `/rfqs/awards` · SRC-04 new (logic from `sealed_bid_mip.py`) · supplier side ← `/supplier/rfqs`
TESTS	RFQ-2231 scenario = 60/40 weighted 400.8; early unseal attempt blocked and audited

M06 Supplier Lifecycle & Performance `Source & Contract · S7, S13`

TABLES	supplier, supplier_category_status, qualification, questionnaire_response, certificate, scorecard, scorecard_period, risk_signal, capa, supplier_user
STATES	Per category: Registered → Under Qualification → Qualified | Conditional → Preferred | Blocked | Exited
APIS	`/v1/suppliers/*`, `/v1/suppliers/{id}/qualifications`, `/scorecards`, `/risk-signals`
EVENTS	supplier.status.changed; consumes goods_receipt.posted, invoice.posted
RULES	Only Qualified/Preferred receive POs for the category (one-time vendor policy aside); expired certificate → Conditional; block stops new POs and flags open ones
SCREENS	SLM-01 ← `/suppliers/360`, `/supplier/[id]` · SLM-02 ← `/suppliers/onboarding`, `/assessments`, `/approvals` · SLM-03 ← `/buyer/suppliers/scorecards` · SLM-04 ← `/risk/*` · SLM-05 ← `/supplier/*` (moves to portal app)
TESTS	Certificate-expiry timer; block-with-open-PO flow; scorecard recompute on events

M07 Contract Lifecycle Management `Source & Contract · S12`

TABLES	contract, contract_line, clause, clause_library, playbook_position, template, obligation, price_validity, consumption, signature_envelope
STATES	Draft → In Negotiation → In Approval → Out for Signature → Active → Expiring → Expired | Terminated | Renewed
APIS	`POST /v1/contracts`, `/activate`, `GET /v1/contracts/{id}/consumption`, `/obligations`
EVENTS	contract.activated, contract.consumption.threshold
RULES	Only executed contracts referenced by POs; PO price = contract ± tolerance; legal + finance above threshold; indexation creates new validity periods
SCREENS	CLM-01 ← `/contracts`, `/contracts/view` · CLM-02 ← `/workbenches` clause redline · CLM-03 ← `/contracts/pending-renewals` · templates ← `/contracts/templates` · CLM-04 new
TESTS	SA-5520 LME reset creates new validity; consumption 80/90/100% alerts

M08 Operational Purchasing `Order to Pay · S6`

TABLES	purchase_order, po_line, po_schedule_line, po_version, po_confirmation, po_change_request, info_record, source_list, transmission
STATES	Draft → In Approval → Released → Transmitted → Confirmed → Partially Received → Received → Invoiced → Closed | Cancelled; any change after confirmation → new version
APIS	POST `/v1/purchase-orders`, `/validate`, `/release`, `/transmit`, `/confirmations`, `/versions`
EVENTS	purchase_order.released, .confirmed, .changed
RULES	DMN-02 touchless; additive risk score; no release to blocked supplier; split detection escalates; deviation reason when price leaves tolerance; SOD-01, SOD-03, SOD-04
SCREENS	PO-01 ← `/purchase-orders` · PO-03 ← `/supplier/orders` + confirmations · PO-04 ← `/workbenches` expediting · PO-02 new `/purchase-orders/[id]`
TESTS	PO-77410 risk 18, touchless; illegal transitions return 409; stale update returns 412

M09 Fulfilment, Goods Receipt & Service Entry `Order to Pay · S7`

TABLES	asn, asn_line, goods_receipt, gr_line, inspection_lot, usage_decision, return_delivery, service_entry, ses_line; inventory stock and movements
STATES	GR: Posted → Quality Hold → Accepted | Partially Rejected | Rejected → Reversed. SES: Draft → Submitted → Approved | Rejected → Revoked
APIS	`POST /v1/asns`, `POST /v1/goods-receipts`, `/reverse`, `POST /v1/service-entries`, `/approve`
EVENTS	asn.received, goods_receipt.posted, quality.usage_decision, service_entry.approved
RULES	Over-delivery above tolerance blocked; quality-managed GR to quality stock; SES approver differs from supplier and PO creator; receipt-less expense only below value
SCREENS	RCV-01 ← `/supplier/shipping/asn`, `/buyer/logistics` · RCV-02 ← `/inventory/receipt` · RCV-03 ← `/workbenches` service entries · RCV-04 new
TESTS	GR-90231: 5,400 received, 5,380 accepted, 20 rejected

M10 Invoice Matching & Finance Integration `Order to Pay + Integration · S8`

TABLES	invoice, invoice_line, tax_line, match_result, invoice_exception, credit_note, posting, payment, payment_block, grir_accrual, integration_message
STATES	Captured → Duplicate Check → Matching → Exception | Matched → Approved → Posted → Paid | Rejected
APIS	`POST /v1/invoices`, `/match`, `/exceptions/{eid}/resolve`, `/post` (Idempotency-Key), `/v1/payments`
EVENTS	invoice.captured, invoice.exception.raised, invoice.posted, payment.completed
RULES	DMN-03; unique + fuzzy duplicate; 3-way needs GR unless ERS; services against SES; posting key tenant + company code + supplier + invoice no.; payment block until exceptions resolved
SCREENS	INV-01 ← `/invoices`, `/finance/invoices` · INV-02 ← `/invoices/3-way-match` · INV-04 ← `/finance/payments`, `/finance/dashboard` · INV-03 new · supplier side ← `/supplier/invoices`
TESTS	INV-TF-8812 → QTY +20 → CN-TF-221 → posted 5100004421; replay returns `"replayed": true`

M11 Savings & Value Realisation `Source & Contract · S14`

TABLES	savings_initiative, savings_ledger_entry, baseline
STATES	Identified → Negotiated → Contracted → Realising → Validated | Lost
APIS	`GET /v1/savings`, `POST /v1/savings/initiatives`, `/validate`
EVENTS	Consumes sourcing.event.awarded, invoice.posted; emits savings.realised
RULES	Realised only from posted invoices; cost avoidance never reduces budget; validator differs from owner (SOD-07)
SCREENS	SAV-01, SAV-02 ← `/savings`
TESTS	SV-0412: negotiated 0.99 Cr/yr, avoidance 0.32 Cr, realised YTD 0.38 Cr

M12 Predictive Intelligence & Agents `Intelligence · S17–S19`

TABLES	agent, agent_version, autonomy_policy, kill_switch, agent_run, recommendation, feedback, model_version, copilot_session, retrieval_chunk (pgvector)
APIS	`POST /v1/copilot/ask`, `POST /v1/agents/{agent}/runs`, `PUT /v1/agents/{agent}/autonomy`, `POST /v1/agents/{agent}/kill-switch`
RULES	Agents act only with their role's permissions; L3/L4 only in configured categories and values; every action logged; answer contract mandatory; numbers from tools, never generated
SCREENS	AIX-01 ← `/command-centre` · AIX-02 ← `/copilot` · AIX-03 ← `/command-centre/ai-agents`
TESTS	Autonomy guard denial cases; cross-tenant retrieval leak test; answer-contract completeness ≥ 99%

M13 Platform, Workflow & Administration `Platform · S0–S3, S20`

TABLES	Org model, currency, fx_rate, factory_calendar, number_range, role, permission, role_assignment, sod_rule, sod_violation, action_log, rule_set, process_def, audit_event, outbox_event, integration_message, idempotency_record, industry_template
APIS	`/v1/admin/*`, `GET /v1/audit/{objectType}/{id}`, `POST /v1/integration/messages/{id}/replay`
RULES	Rule and workflow versions are effective-dated; in-flight documents keep their version
SCREENS	ADM-01 ← `/admin/settings` · ADM-02 ← `/admin/governance` · ADM-03 ← `/bpmn-flows` (bpmn-js) · ADM-04 ← `/admin/users` · ADM-06 ← `/admin` · templates ← `/industry-templates` · ADM-05 new
TESTS	RLS suite; audit chain tamper detection; rule version pinning on in-flight documents

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9. Principle traceability

Where each mandatory principle is enforced, when it lands, and how it is proven.

ID	PRINCIPLE	ENFORCEMENT POINT	SPRINT	PROOF
P01	Traceable purpose	`purpose_code NOT NULL` on PR, PO, contract; submit guard	S4	API test: submit without purpose → 422 `RULE_VIOLATION`
P02	Two demand origins	Origin enum; trigger rejects updates	S4	DB test: update origin fails
P03	Reuse before create	Guided-buying search service ordering; agent uses same service	S5	E2E: free text offered only after no contract/catalogue/material hit
P04	No master pollution	CHECK constraint; no code path creates material from PR	S4	Unit + DB constraint test

ID	PRINCIPLE	ENFORCEMENT POINT	SPRINT	PROOF
P05	Governed material creation	F9 with duplicate gateway at 0.92	S5	Workflow test with Zeebe test engine
P06	Evidence-based sourcing	Publish guards (approved requirement, qualified invitees, weights 100)	S10	API tests per guard
P07	Inheritance into PO	PO creation copies references; `deviation_reason` CHECK	S6, S12	RFQ-2231 → SA-5520 → PO-77410 test
P08	Purchase-type rules	Purchase-type matrix as rule set driving GR, valuation, match	S4–S8	Parameterised tests across 7 types
P09	Approvals & SoD	Camunda tasks + SoD engine at command time	S2–S3	SoD suite SOD-01…07 for humans and agents
P10	Complete lineage	`document_link` written on every conversion	S3 onward	Lineage test from PO line to MRP run
P11	Explainable AI	Answer-contract schema validation; run log	S17–S18	Contract completeness eval
P12	Three kinds of savings	Ledger from invoice.posted only	S14	SV-0412 test
P13	Exactly-once posting	Outbox + `integration_message` unique key + query-before-retry	S3, S8	10,000-invoice soak with fault injection
P14	Configurable everything	Effective-dated rule sets by scope	S3	Version pinning test; CI check for hardcoded thresholds

10. SoD implementation

The current `sod_service.py` uses different rule IDs from the blueprint. In S2 the IDs are re-aligned to this table, which is the default tenant rule set.

RULE	ACTION A	CONFLICTS WITH ACTION B	RISK	CHECKED AT
SOD-01	Create supplier master	Create PO for same supplier	High	PO create / release
SOD-02	Change supplier bank details	Approve payment / post invoice	Critical	Invoice approve, post, payment run
SOD-03	Create PO	Post GR for same PO	High	GR post
SOD-04	Create PO	Approve invoice exception for same PO	High	Exception resolve
SOD-05	Award sourcing event	Score same event as sole evaluator	Medium	Award approve
SOD-06	Request material	Approve same material as steward	Medium	Steward decision
SOD-07	Initiate savings	Validate same savings	High	Savings validate

Each check reads `action_log` for the object and its related objects, applies to agents running under a role, and either blocks (409 `SOD_CONFLICT`) or routes to a mitigating approval as the tenant configures. Every outcome is written to `sod_violation` and the audit chain.

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11. Security & privacy

CONTROL	IMPLEMENTATION	SPRINT
Identity	Keycloak OIDC/SAML federation; MFA for approvers, finance, admins; supplier users by invitation with MFA	S2

CONTROL	IMPLEMENTATION	SPRINT
Authorisation	RBAC scopes + ABAC (company code, plant, category, value band) per request; CI check that every route declares a scope	S2
Tenant isolation	RLS on every table, non-owner DB role, tenant from token only	S1
Data protection	TLS 1.3; AES-256 at rest; field-level encryption for bank accounts and tax IDs with per-tenant Cloud KMS keys; masked display	S1, S7
Bank-change fraud	Dual finance approval, call-back, cooling-off, alert on change + invoice within 30 days	S7
Audit	Append-only, hash-chained, 10-year configurable retention, nightly verification	S2
Application security	OWASP ASVS L2; Semgrep SAST, ZAP DAST, pip-audit/npm audit, Trivy image scan, SBOM, signed images; secrets only in Secret Manager	S0 onward
Privacy (DPDP Act)	Purpose limitation, India residency default, retention and deletion workflows, PII redaction at LLM gateway	S17, S24
Threat modelling	STRIDE review per module before its first sprint; pen test in S24	Ongoing

12. NFRs & operations

ATTRIBUTE	TARGET	HOW	VERIFIED BY
Availability	99.9% transactional · 99.5% analytics	Regional GKE, HA Cloud SQL, multi-zone broker	SLO dashboards from S9
Performance	P95 < 400 ms · PR form < 2 s	Indexed queries, read models, Redis caching, cursor pagination (max 200)	k6 per release

ATTRIBUTE	TARGET	HOW	VERIFIED BY
Spend cube	< 3 s on 50 M lines	ClickHouse with CDC refresh < 5 min	ClickHouse benchmark S14, S24
MRP	50,000 materials < 10 min	Partitioned workers, vectorised netting	S16 benchmark
Scale	10,000 concurrent users/tenant · 5 M invoices/yr	Horizontal API pods, broker partitioning by tenant	S24 load test
Recovery	RPO 15 min · RTO 4 h	PITR, cross-region replica, Terraform rebuild	S24 DR drill
Accessibility	WCAG 2.1 AA	Design system components, jsx-a11y lint, axe in CI	Audit S24

OBSERVABILITY

- OpenTelemetry traces across web, API, workers, Zeebe and connectors; `traceId` in every error response.
- Golden signals per module; business SLOs for PR approval latency, touchless rate, first-pass match rate and posting success rate.
- Runbooks for dead-lettered postings, workflow incidents, MRP run failure and model drift; on-call rota from S9.
- Blue/green deployments; migrations stay backward-compatible for one release (expand, migrate, contract).

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13. CI/CD & environments

PIPELINE

1. **Pull request:** ruff, mypy (strict on `modules/*/states.py`, `rules.py`), pytest unit and API, eslint, `tsc`, Vitest, OpenAPI breaking-change diff, Avro compatibility, gitleaks, Semgrep, dependency audit.

2. **Merge to main:** build images, Trivy scan, SBOM, cosign signature, deploy to dev, integration tests with Testcontainers and Zeebe, migration upgrade/downgrade check.
3. **Nightly:** RLS and SoD suites, Camunda process tests, ZAP against test environment, Pact verification.
4. **Release:** Playwright E2E, k6, change approval, blue/green to production.

ENVIRONMENTS

ENV	PURPOSE
local	docker-compose dev stack
dev	Auto-deploy from main
test	Nightly suites, ERP sandboxes
staging	Release candidates, UAT, anonymised pilot data
prod	asia-south1, DR asia-south2

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14. Testing & release gates

LAYER	SCOPE	TOOLING	GATE
Unit	Netting, match, TCO, risk, rules, state machines	Pytest, Hypothesis, Vitest	85% coverage on domain logic
Contract	Service APIs and event schemas	Pact, schema registry compatibility	Must pass to merge
Integration	Workflow + modules + DB with RLS	Testcontainers, Zeebe test engine	Nightly

LAYER	SCOPE	TOOLING	GATE
E2E	E2E-01…n through UI	Playwright	Release gate
Security	RLS isolation, SoD, authZ, OWASP	Custom suites, ZAP, external pen test	Release gate
Performance	NFR targets	k6, ClickHouse benchmarks	Per release
AI evaluation	Accuracy, MAPE, answer-contract completeness	Offline eval sets + human review	Before any autonomy increase

The worked example (Axion Mobility: MR-1043 → REQ-2311 → RFQ-2231 → SA-5520 → PO-77410 → ASN-61208 → GR-90231 → INV-TF-8812 → SV-0412) is implemented as one executable scenario that grows each phase and must stay green on every release.

15. Data migration & cutover

OBJECT	SOURCE	APPROACH	VALIDATION	REHEARSALS
Suppliers & materials	ERP masters	Initial load + delta sync	Counts, duplicate scan, bank verification sample	S9, S14, S24
Open contracts	ERP, shared drives	Template load; AI clause extraction with review	Value and validity totals	S14, S24
Open POs	ERP	Load with ERP number as external reference	Open value per supplier matches	S9, S24
Spend history	24–36 months PO + invoice	Bulk load to ClickHouse; classification run	Ties to GL ±0.5%	S14, S24
Open invoices	ERP AP	Read-only references; new invoices through platform	AP ageing matches	S24

OBJECT	SOURCE	APPROACH	VALIDATION	REHEARSALS
Current demo data	Postgres / `capability_twin.db`	One-off script in S1 into new schema; then discarded	Row counts	S1

16. Existing code disposition

What happens to each part of the current codebase.

CODE	ACTION	DETAIL
`services/rules_engine.py`	REFACTOR	Keep logic; thresholds read from rule sets (S3)
`services/sod_service.py`	REFACTOR	Move to DB, re-align rule IDs, remove seeded fake violations (S2)
`services/lineage_service.py`	REFACTOR	BFS over `document_link` (S3)
`services/outbox_service.py`	REPLACE	Transactional outbox + relay + connector worker (S3)
`routers/purchasing_p2p.py`	REBUILD	Logic reused in requisition, order, receiving, payables modules (S4–S8)
`routers/procurement.py`, `finance.py`, `actions.py`	MIGRATE, THEN RETIRE	Data migrated; frozen in S1; removed once pages use `/v1` (S9)
`sourcing_clm_savings.py`, `contracts.py`, `sealed_bid_mip.py`	REBUILD	sourcing, contract, savings modules (S10–S14)
`planning_mdm.py`, `mdm_governance.py`	REBUILD	mdm (S5, S7), planning (S15)
`unified_approvals.py`, `invoice_exceptions.py`	REBUILD	Camunda task inbox (S3); payables exceptions (S8)
`admin.py`, `enterprise_admin.py`, `tenant.py`	REBUILD	admin module on Keycloak and DB (S1–S3)

CODE	ACTION	DETAIL
`copilot.py`, `spend_cube_ai_agents.py`, `core/agent_graph.py`	REPLACE	ai module (S17–S18)
`routers/inventory.py`	KEEP, HARDEN	Add auth, RLS, money types; feeds MRP on-hand (S7)
Supplier profile, onboarding, evidence, scoring, twin, development, DFSC, ESG, network, discovery	KEEP IN TWIN CONTEXT	Persist in-memory stores; expose §12.3 interface (S13)
`iot.py`, `blockchain.py`, `simulation.py`, `analytics.py`, `analytics_hub.py`, geospatial page	FEATURE-FLAG OFF	Not in the blueprint; remove random data; product owner decides scope after GA
`app/api/*` (unregistered), `telemetry.py`, `inventory.py.bak`, `core/db.py`	DELETE	S0
`test_*.py`, `verify_*.py`	PORT	Into pytest suites (S0–S1)
Frontend pages	REFACTOR	Keep layout and flows; move to design system and typed client; replace inline seed arrays; split pages over 600 lines
`components/layout/Shell.tsx`	REFACTOR	Navigation from user scopes, grouped Decide · Plan & Request · Source & Contract · Order to Pay · Reference; lifecycle strip; Copilot in header

17. Risks

RISK	IMPACT	MITIGATION	OWNER
Team smaller than six squads	Dates slip up to 2×	Keep phase order; cut industry packs to manufacturing + general first	Delivery manager

RISK	IMPACT	MITIGATION	OWNER
Camunda 8 licence or learning curve	Workflow delays from S3	Confirm licence in S0; spike in S1; Camunda SaaS as fallback	Architect
Late ERP sandbox access	S8 posting blocked	Request in S0; WireMock contract stubs from SAP API specs	Integration lead
Frontend breaks during legacy API retirement	Demo regressions	Freeze legacy API in S1; migrate pages module by module behind feature flags	Order to Pay lead
Twin features compete for capacity	Core PIC scope slips	Twin work limited to the §12.3 interface until GA	Product owner
Pilot data quality	Migration rework	Three rehearsals; duplicate and bank checks before load	Integration lead
LLM accuracy, cost or data leakage	Trust and compliance	Tool-computed numbers, permission-filtered retrieval, eval gates, gateway budgets	Intelligence lead
MIP award solve times	Slow SRC-04	Bound supplier count; CP-SAT time limits; cache scenarios	Source & Contract lead

18. Definition of done

- Acceptance criteria and linked workbook test cases pass; the screen matches its blueprint ID and states.
- Every command checks scope and ABAC, runs SoD where a rule applies, writes an audit event and, where relevant, an outbox event in the same transaction.
- State transitions go through the module's state machine; illegal ones return 409; stale writes return 412.
- Thresholds live in rule sets; no hardcoded values (CI check).
- Money in minor units with currency; dates in UTC.

- OpenAPI and Avro schemas updated; idempotency, concurrency and `problem+json` errors handled.
- RLS test added for any new table.
- Frontend uses the typed client and design system; keyboard-complete; axe passes; loading, empty and error states designed.
- Runbook and dashboards updated for any new failure mode.

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19. First ten working days

Sprint 0, 26 October to 6 November 2026.

DAY	WORK	OWNER
1	Git repository, branch protection, baseline commit of current code	Platform lead
1–2	Fix requirements, delete dead code, remove backdoor and hardcoded secrets, add settings module	Platform
2–3	Dev stack compose file with Postgres 16, Redpanda, Keycloak, Camunda, Redis, OTel	DevOps
2–4	Pytest harness with Testcontainers; port verify scripts	QA leads
3–4	CI v1 pipeline	DevOps
3–5	Frontend API URL env var, Providers fix, legacy API freeze list	Order to Pay
4–6	Interim guards: invoice double-pay, PO state endpoint locked down	Order to Pay
5–7	ADRs DD-01 to DD-11; Camunda licence; ERP sandbox requests	Architect
6–9	Design system inventory from existing pages; component specs	UX + Plan & Request

DAY	WORK	OWNER
8–10	S1 backlog groomed: schema baseline, RLS, org model, problem+json, idempotency	Platform lead